

Project Report: Java Blockchain Simulator

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1. Objective

The goal of this project was to understand how a blockchain maintains data integrity and security. I built a desktop application using **Java Swing** that simulates the creation of blocks, the mining process (Proof of Work), and the detection of data tampering.

2. Core Features

- **SHA-256 Hashing:** Used the MessageDigest library to generate unique digital fingerprints for every block.
- **Proof of Work (Mining):** Implemented a "mining" mechanism where the system searches for a hash starting with a specific number of zeros (Difficulty: 4).
- **Chain Integrity:** Each block stores the hash of the previous block, creating a linked chain.
- **Tamper Simulation:** Included a specific feature to manually alter block data to show how the system detects an "Invalid" chain immediately.

3. Project Structure

I divided the code into three main files to keep it organized:

- **StringUtil.java:** A utility class to handle the complex SHA-256 math.
- **Block.java:** The blueprint for a single block, containing the data, hash, and mining logic.
- **BlockchainGUI.java:** The main interface built with Swing that lets the user interact with the blockchain visually.

4. Technical Challenges Overcome

- **Handling Hex Strings:** Converting raw byte arrays from the SHA-256 process into readable hexadecimal strings was tricky, but I solved it using a StringBuilder in the StringUtil class.
- **GUI Responsiveness:** Ensuring the JTextArea updated in real-time as blocks were mined.

- **Tamper Logic:** I had to implement a setData() method to manually change a block's private data to demonstrate that the calculateHash() would no longer match the stored hash.

5. Results (Postman & GUI Testing)

- **Mining:** Confirmed that blocks are successfully added only after the "Mine Block" button finishes the Proof of Work.
- **Validation:** Before tampering, the system returns "Chain is Valid!".
- **Tamper Test:** After using the "Tamper Block" button to change index 0's data, clicking "Check Validity" correctly triggered the **"Chain is INVALID!"** warning.



6. Conclusion

This first project gave me a strong foundation in cryptography and data structures. It perfectly demonstrates how blockchain technology prevents unauthorized data changes through linked hashing and computational difficulty.

Project Source Code The complete source code, including the project structure and commit history, is available on my GitHub repository:

https://github.com/Kanish20X/BlockChain_Simulator